Machine Learning

Support Vector Machine

Dong-Kyu Chae

PI of the Data Intelligence Lab @HYU
Department of Computer Science
Hanyang University

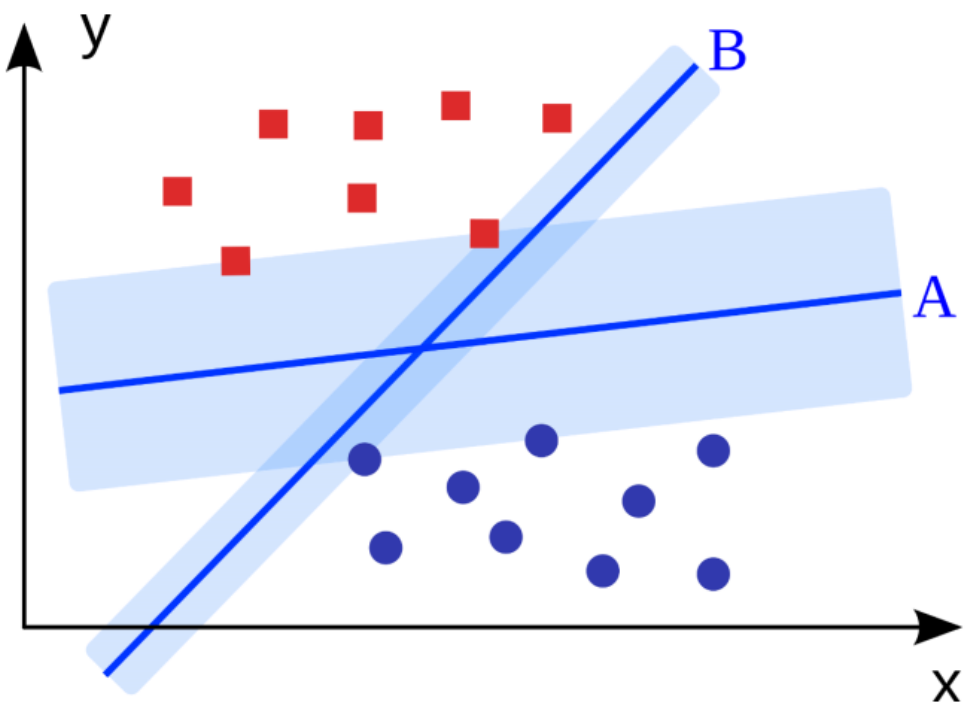
Data
Intelligence
LAB

Contents

- **What is SVM ?**
- **Objective function of SVM**
- **Optimization with constraints**
  - Lagrange multiplier
  - Application to SVM
- **Kernel trick**
  - Application to SVM
- **Soft margin SVM**

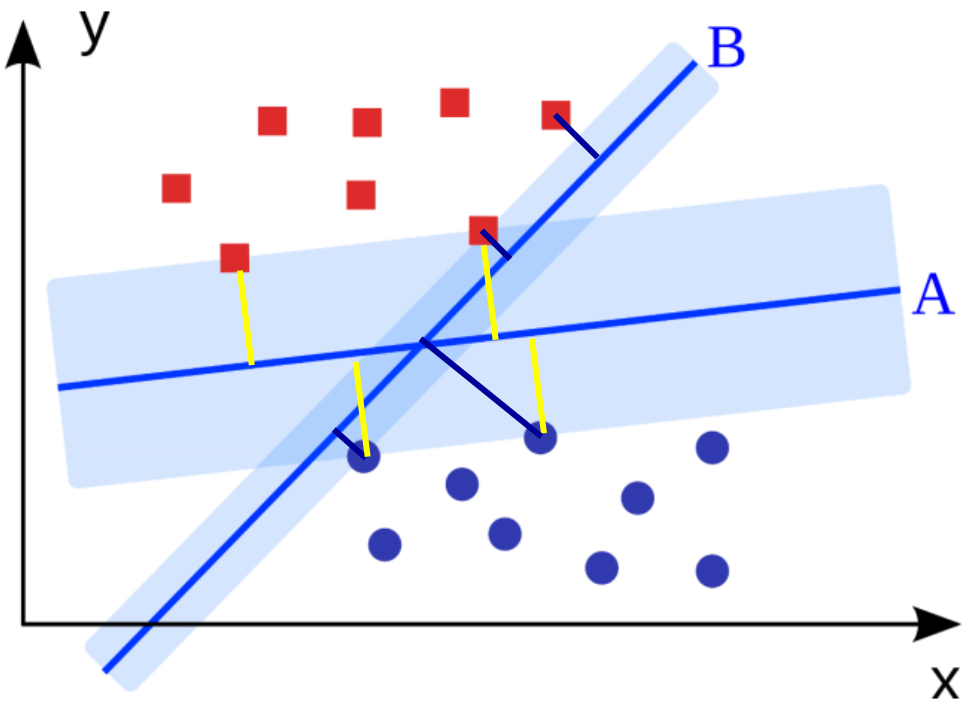
Introduction to SVM

- **Which one is a good decision boundary?**
  - Assumption: our data is linearly (and perfectly) separable
  - Both A and B provides 100% training accuracy, and their model complexity are the same
  - But which one looks better? And why?



Introduction to SVM

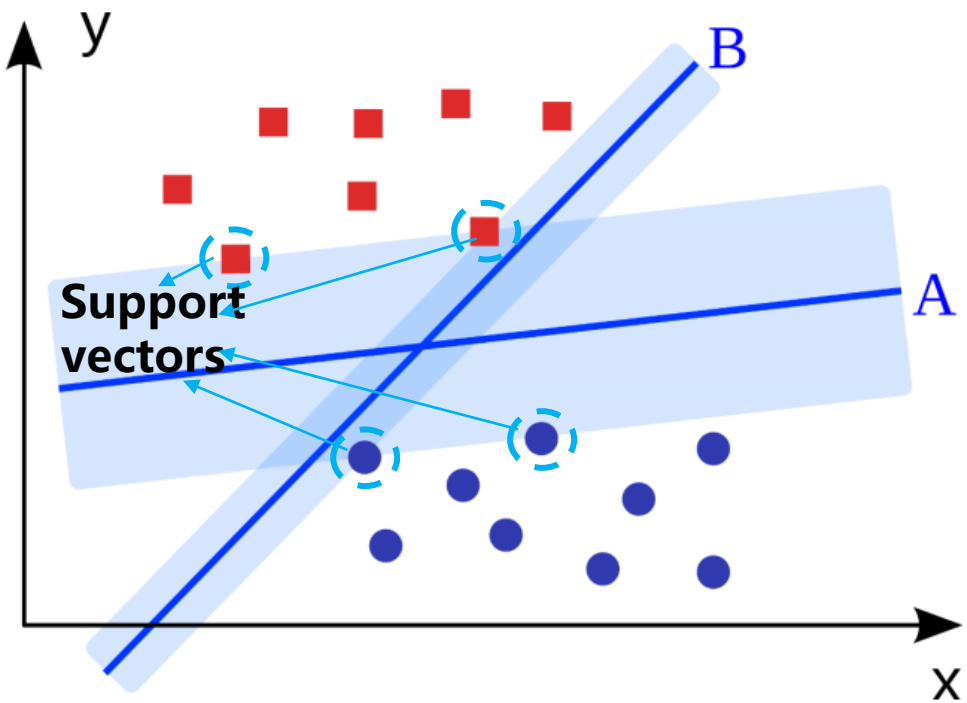
- **A looks better than B**
  - Why? A leaves more **margin** on either class of data
  - Margin: Perpendicular distance from the closest point to the decision boundary
  - Higher margin will provide better **generalization performance**!



Introduction to SVM

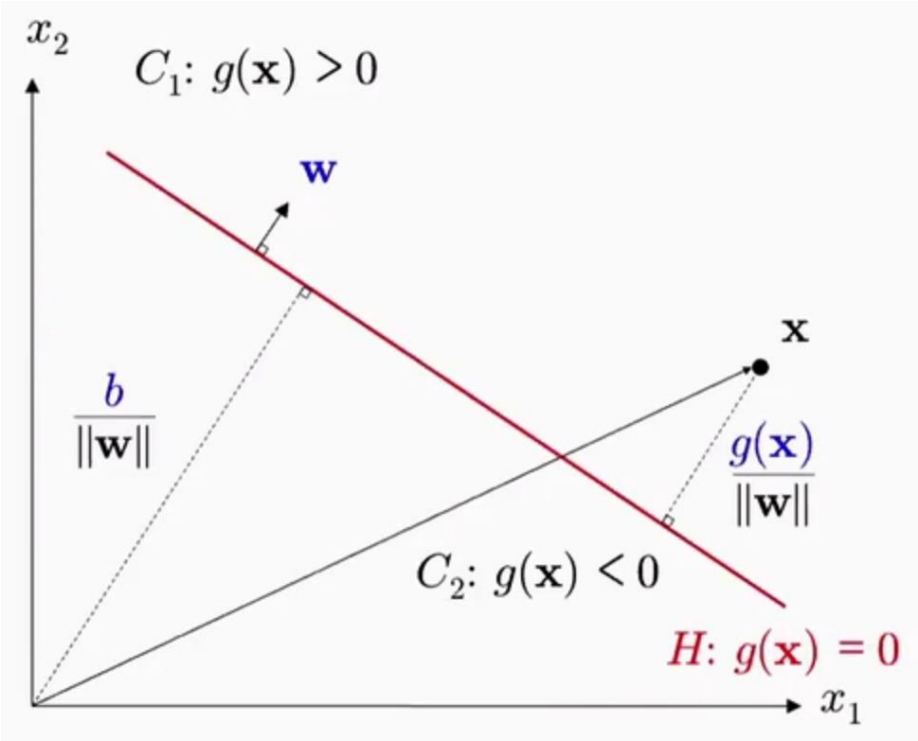
Idea of SVM

- Find a decision boundary (in general, it is a hyperplane) that leaves **maximum margin** from both classes
- We call this hyperplane as "**Support Vector Machine**"
  - Support vector: a data instance with the minimum margin



Introduction to SVM

- **Some mathematical points**
  - $\mathbf{x}$: data (vector), $g(\mathbf{x}) = 0$: a decision boundary
  - $\mathbf{w}$: weight vector & normal to the hyperplane $g(\mathbf{x}) = 0$
    - $\mathbf{w}^T\mathbf{x}_1 + b = \mathbf{w}^T\mathbf{x}_2 + b \;\; => \mathbf{w}^T(\mathbf{x}_1 - \mathbf{x}_2) = \mathbf{0}$
  - b: bias
  - If $\mathbf{x}_\mathrm{p}$ is on the decision boundary, $g(\mathbf{x}_\mathrm{p}) = 0$
  - If $\mathbf{x}_a$ is above the boundary, $g(\mathbf{x}_a) > 0$
  - If $\mathbf{x}_b$ is below the boundary, $g(\mathbf{x}_b) < 0$



Introduction to SVM

***note:** $||\mathbf{w}|| = \sqrt{w_1^2 + w_2^2 + \cdots + w_d^2}$

Some mathematical points

- Consider an arbitrary data point $\mathbf{x}$
  - The signed distance from $\mathbf{x}$ to the decision boundary, $r = \frac{g(\mathbf{x})}{||\mathbf{w}||}$

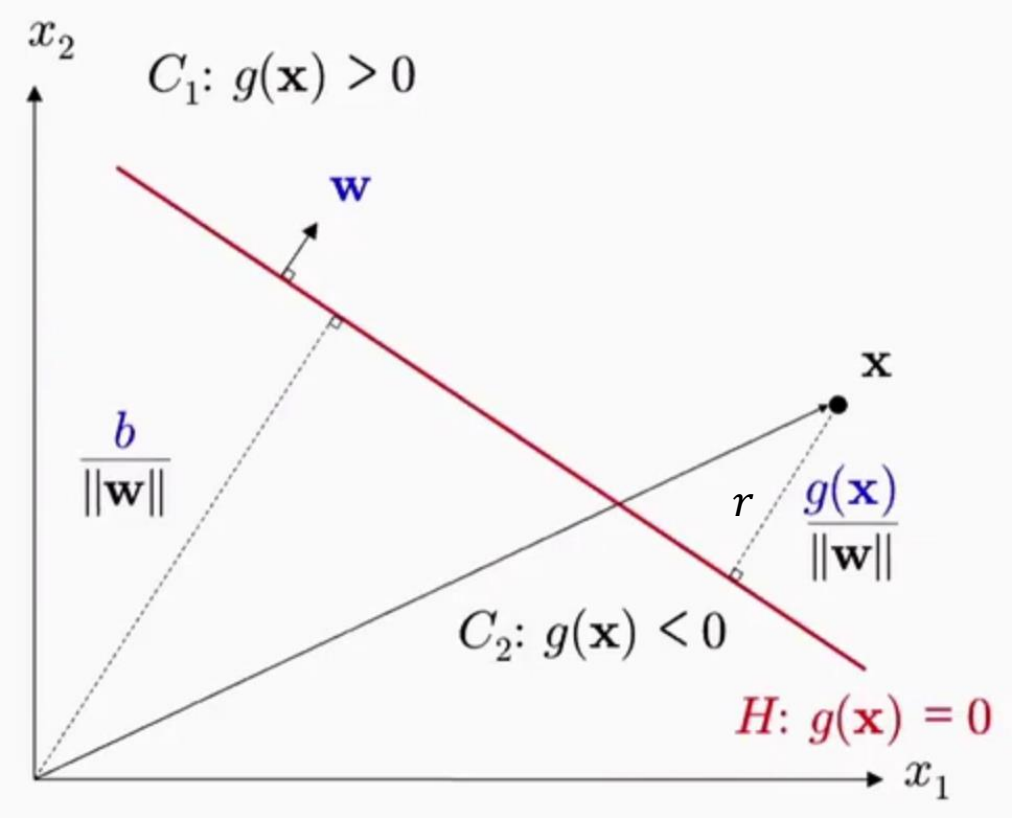
$$\mathbf{x} = \mathbf{x}_\mathrm{p} + r\frac{\mathbf{w}}{||\mathbf{w}||}, \qquad g(\mathbf{x}_\mathrm{p}) = 0$$

$$g(\mathbf{x}) = \mathbf{w}^\mathsf{T}\mathbf{x} + b = \mathbf{w}^\mathsf{T}\left(\mathbf{x}_\mathrm{p} + r\frac{\mathbf{w}}{||\mathbf{w}||}\right) + b$$

$$= \underline{\mathbf{w}^\mathsf{T}\mathbf{x}_\mathrm{p} + b} + r\frac{\mathbf{w}^\mathsf{T}\mathbf{w}}{||\mathbf{w}||}$$

$$g(\mathbf{x}) = r||\mathbf{w}||$$

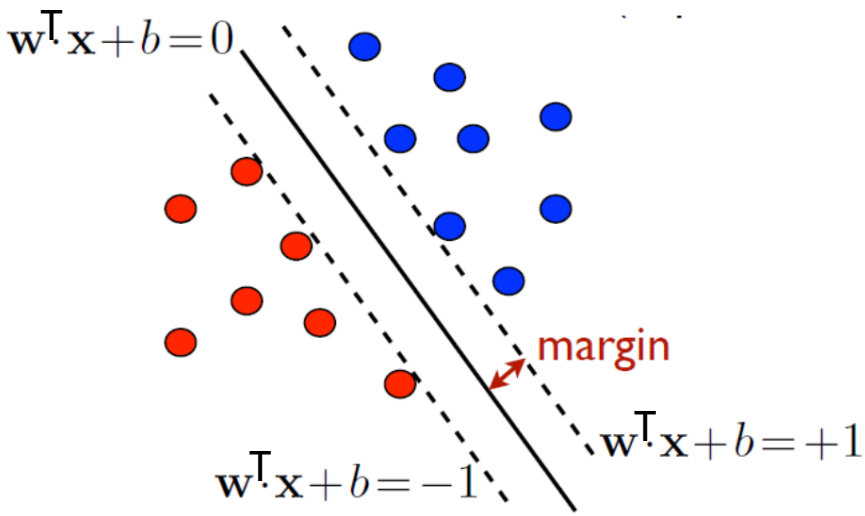
$$r = \frac{g(\mathbf{x})}{||\mathbf{w}||}$$



Objective Function

- **Objective: Find a hyperplane that maximizes the margin!**
  - Margin: distance between the **support vectors** and the **hyperplane**
  - Objective: maximizing 2xMargin

- **Formally,**
  - $\mathbf{x}_-$, $\mathbf{x}_+$: support vectors associated with each class
  - Margin $r = \dfrac{g(\mathbf{x}_-)}{||\mathbf{w}||} = \dfrac{g(\mathbf{x}_+)}{||\mathbf{w}||}$



- **First step:** we scale $\mathbf{w}$ and b to make the following results:
  - $g(\mathbf{x}_+) = +1$
  - $g(\mathbf{x}_-) = -1$

Objective Function

***note:** $||\mathbf{w}|| = \sqrt{w_1^2 + w_2^2 + \cdots + w_d^2}$

- **Then the margin becomes:** $r = \frac{\pm 1}{||\mathbf{w}||}$
  - 2xMargin = $2|r| = \frac{2}{||\mathbf{w}||}$
  - Therefore, we want to **maximize** $\frac{2}{||\mathbf{w}||}$

    => equivalent to **minimize** $||\mathbf{w}||$

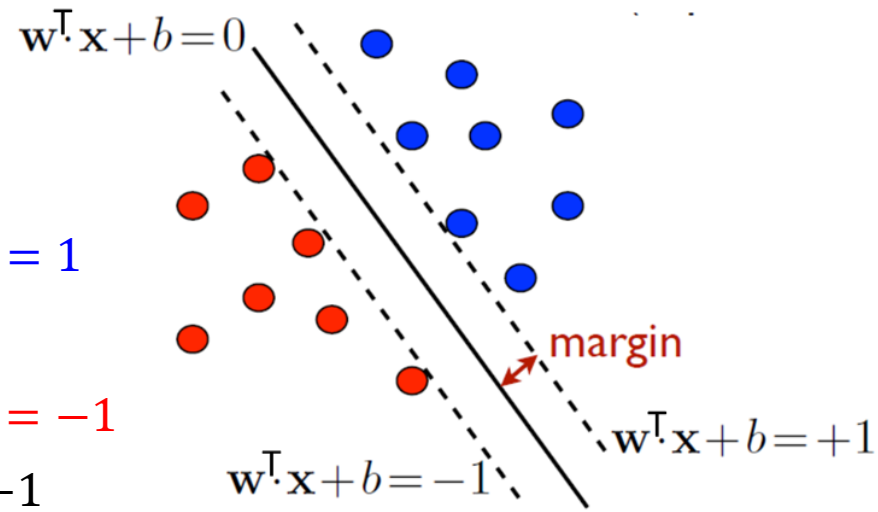
    => equivalent to **minimize** $\frac{1}{2}||\mathbf{w}||^2$
  - Therefore, the cost function becomes: $\boxed{\mathbf{w}^* = \min \frac{1}{2}||\mathbf{w}||^2}$
  - But, its solution will be…

$$\mathbf{w}^* = [w_1^*, w_2^*, \dots, w_d^*] = [0, 0, \dots, 0]$$

What we need additionally?

=> **constraints!**

Objective Function



- **Constraints**
  - For data with class "+1" => $y_{+1} = 1$
    - $g(\mathbf{x}_{+1}) \geq 1$ or $\mathbf{w}^{\mathbf{T}}\mathbf{x}_{+1} + b \geq 1$
  - For data with class "-1" => $y_{-1} = -1$
    - $g(\mathbf{x}_{-1}) \leq -1$ or $\mathbf{w}^{\mathbf{T}}\mathbf{x}_{-1} + b \leq -1$
- **Combined constraints**
  - $y_i(\mathbf{w}^{\mathbf{T}}\mathbf{x}_i + b) \geq 1$ for **all data** from $(\mathbf{x}_1, y_1)$ to $(\mathbf{x}_N, y_N)$
  - Multiple linear inequality constraints (# of data instances)
- **Finally, the objective function is:**

$$\min \quad \frac{1}{2}\|\mathbf{w}\|^2$$

$$s.t. \quad y_i(\mathbf{w}^{\mathrm{T}}\mathbf{x}_i + b) \geq 1, \quad \forall i \in [\mathbf{1}, \boldsymbol{N}]$$

Optimization with Constraints

- **How to solve this optimization problem?**

$$\min \quad \frac{1}{2}\|\mathbf{w}\|^2$$

$$s.t. \quad y_i(\mathbf{w}^{\mathrm{T}}\mathbf{x}_i + b) \geq 1, \quad \forall i \in [\mathbf{1}, \boldsymbol{N}]$$

- A constrained optimization problem
  - There are N constraints in total
- We can solve this problem by using the **Lagrange multiplier**
  1. Define Lagrange multiplier $\boldsymbol{\alpha} = < a_1, a_2, \ldots, a_N >$ (they are all ≥ 0)
  2. Define Lagrange (objective) function, where all the constraints are eliminated:

$$L_p(\mathbf{w}, b, \alpha) = \underbrace{f(\mathbf{w})}_{\text{objective}} - \sum_{i=1}^{N} \alpha_i \underbrace{g_i(\mathbf{w}, b)}_{\text{constraints}}$$

  3. Define the **dual problem**: $\underset{\alpha}{\operatorname{argmax}}\{ \underset{\mathbf{w},\, b}{\operatorname{argmin}}\{ L_p(\mathbf{w}, b, \alpha) \} \}$, and solve it!

Optimization with Constraints

- **Lagrange multiplier**
  - The method of Lagrange multipliers is a strategy for optimizing a function $f(x)$ subject to constraints, each $g_i(x) = 0$
  - This method converts a constrained problem into an unconstrained form by multiplying the Lagrange multipliers to each constraints:

$$L(\mathbf{w}, b, \alpha) = f(\mathbf{w}) - \sum_{i=1}^{N} \alpha_i g_i(\mathbf{w}, b)\text{, each } a_i \neq 0$$

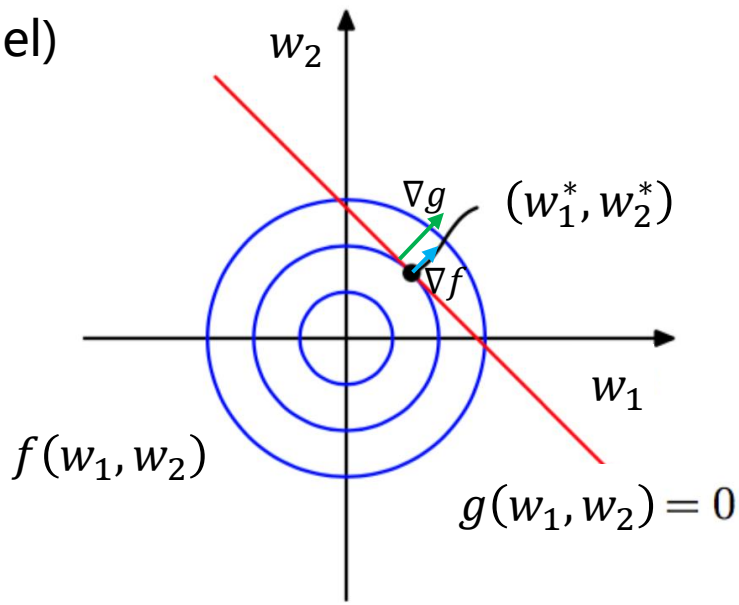
- **Key idea:**
  - $\nabla f(\mathbf{w}^*) = \alpha \cdot \nabla g(\mathbf{w}^*)$, which means that the two gradient vectors $\boldsymbol{\nabla f}$ and $\boldsymbol{\nabla g}$ are **parallel** (or, anti-parallel) on an optimal point! (the magnitudes of the two vectors are generally not equal).

Lagrange Multiplier

- **Why** $\nabla f(\mathbf{w}^*) = \alpha \cdot \nabla g(\mathbf{w}^*)$ **[1]**
  - Let's think about a simple problem:

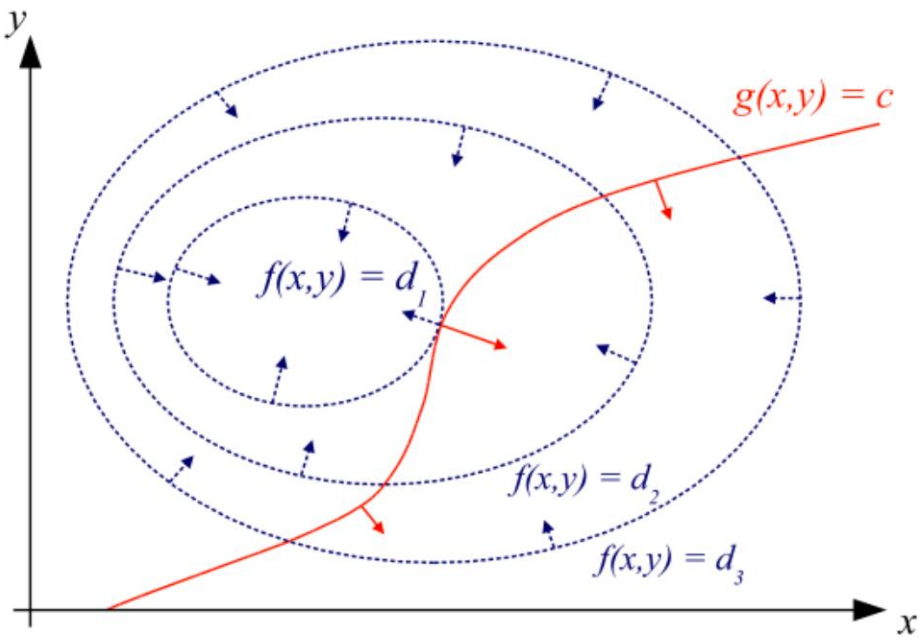
$$\text{Minimize } f(w_1, w_2) = w_1^2 + w_2^2$$
$$\text{subject to } g(w_1, w_2) = y_1 \cdot (x_1 w_1 + x_2 w_2 + b) = 0$$

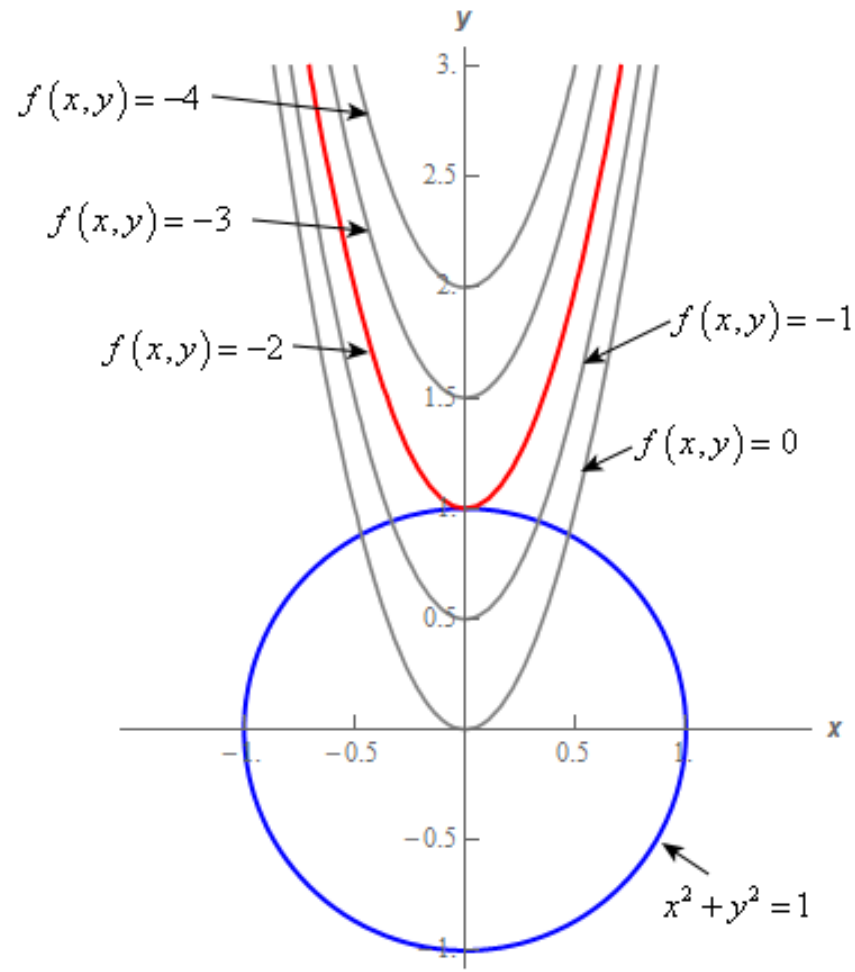
  - In general, the solution exists on the tangent point (접점) among the two functions, and on that point, the two gradient vectors $\boldsymbol{\nabla f}$ and $\boldsymbol{\nabla g}$ are parallel (or, anti-parallel)



Lagrange Multiplier

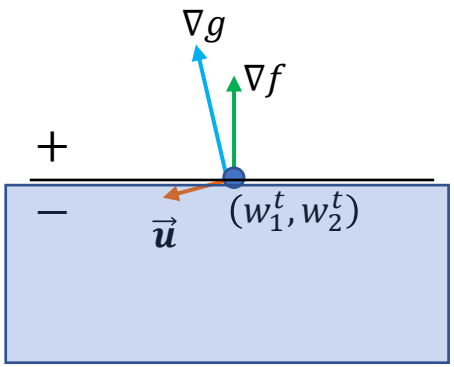
- **Why** $\nabla f(\mathbf{w}^*) = \alpha \cdot \nabla g(\mathbf{w}^*)$ **[1]**
  - Another examples





Lagrange Multiplier

- **Why** $\nabla f(\mathbf{w}^*) = \alpha \cdot \nabla g(\mathbf{w}^*)$ **[2]**
  - Let's think about moving from $\mathbf{w}^t$ to $\mathbf{w}^{t+1}$ following a direction $\vec{\boldsymbol{u}}$
  - In order to satisfy the constraint, the directional derivative $\boldsymbol{\nabla g} \cdot \vec{\boldsymbol{u}}$ should be 0
  - To minimize f, $\boldsymbol{\nabla f} \cdot \vec{\boldsymbol{u}} < 0$
  - However, if $\mathbf{w}^t$ is an optimal point, the directional derivative $\boldsymbol{\nabla f} \cdot \vec{\boldsymbol{u}} = 0$
  - To satisfy both $\boldsymbol{\nabla f} \cdot \vec{\boldsymbol{u}} = \boldsymbol{\nabla g} \cdot \vec{\boldsymbol{u}} = \mathbf{0}$, $\boldsymbol{\nabla f}$ and $\boldsymbol{\nabla g}$ must be parallel (or, anti-parallel) on the optimal point



Lagrange Multiplier

1. **Define the Lagrangian function as:**

$$L(w, \alpha) = f(w) - \alpha g(w)$$

2. **To obtain optimal $\mathbf{w}^*$ and α, solve:**

$$\nabla_{w,\alpha} L = \begin{bmatrix} \frac{\partial L}{\partial w} & \frac{\partial L}{\partial \alpha} \end{bmatrix} = 0$$

3. **Then you can obtain the following equations:**

$$\frac{\partial L}{\partial w} = \nabla f(\mathbf{w}^*) - \alpha \nabla g(\mathbf{w}^*) = 0$$

$$\frac{\partial L}{\partial \alpha} = -g(\mathbf{w}^*) = 0$$

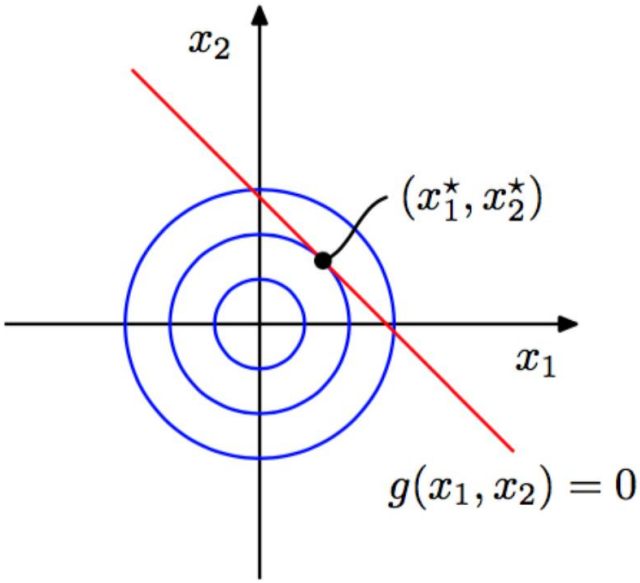
$$\Rightarrow \begin{pmatrix} \frac{\partial f}{\partial w_1} - \alpha \frac{\partial g}{\partial w_1} = 0 \\ \frac{\partial f}{\partial w_2} - \alpha \frac{\partial g}{\partial w_2} = 0 \\ \vdots \\ \frac{\partial f}{\partial w_d} - \alpha \frac{\partial g}{\partial w_d} = 0 \end{pmatrix}$$

Lagrange Multiplier Example

- **Solve the following optimization problem:**

max $f(x_1, x_2) = 1 - x_1^2 - x_2^2$

subject to $g(x_1, x_2) = x_1 + x_2 - 1 = 0$



- **Largrangian function:**

$$L(x, \alpha) = x_1^2 + x_2^2 - 1 - \alpha(x_1 + x_2 - 1)$$

Lagrange Multiplier Example

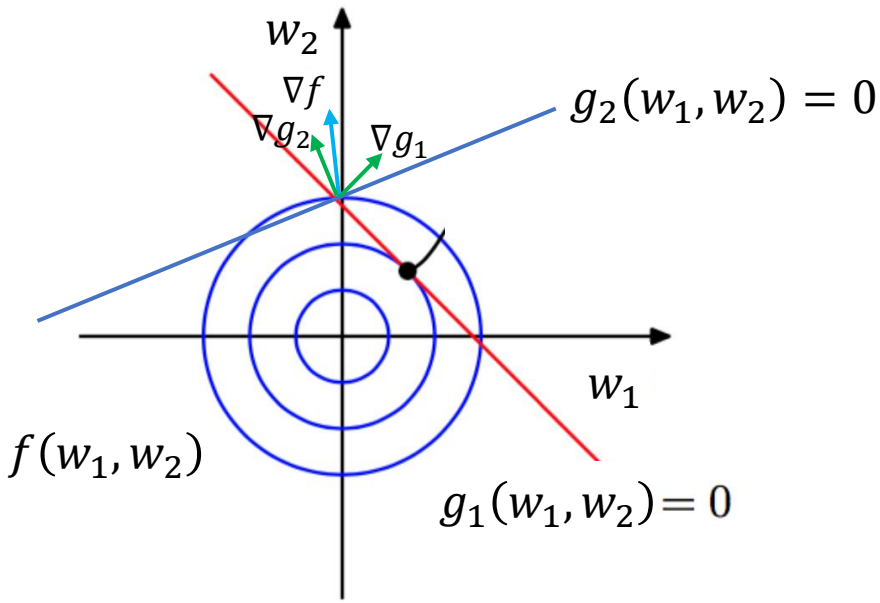
- **Differenting** $L(x, \lambda)$ w.r.t. x_1, x_2**, and** α **and set the results to 0, we obtain**

$$2x_1 - \alpha = 0$$
$$2x_2 - \alpha = 0$$
$$x_1 + x_2 - 1 = 0$$

- **The solution is** $(x_1^*, x_2^*) = \left(\frac{1}{2}, \frac{1}{2}\right)$**, and the corresponding** $\alpha = 1$**. As** $\alpha > 0$**, the gradients of** $f(x_1, x_2)$ **and** $g(x_1, x_2)$ **are parallel at** (x_1^*, x_2^*)**.**

Lagrange Multiplier: Multiple Constraints

- **Suppose that there are multiple equality constraints**



- **There is the following relationship between the gradient vectors on the optimal point:**

$$\nabla f(\mathbf{w}^*) = \alpha_1 \cdot \nabla g_1(\mathbf{w}^*) + \alpha_2 \cdot \nabla g_2(\mathbf{w}^*) + \cdots + \alpha_N \cdot \nabla g_N(\mathbf{w}^*)$$

Lagrange Multiplier: Multiple Constraints

- **If there are N constraints, there are scalars $\alpha_1, \alpha_2, \dots, \alpha_N$ such that**

$$\nabla f(\mathbf{w}^*) = \sum_{k=1}^{N} \alpha_k \nabla g_k(\mathbf{w}^*) \Leftrightarrow \nabla f(\mathbf{w}^*) - \sum_{k=1}^{N} \alpha_k \nabla g_k(\mathbf{w}^*) = 0$$

- **Then the Lagrange function can be generalized to**

$$L(\mathbf{w}, \boldsymbol{\alpha}) = f(\mathbf{w}) - \sum_{k=1}^{N} \alpha_k g_k(\mathbf{w})$$

- **Then, solve:**

$$\nabla_{\mathbf{w},\boldsymbol{\alpha}} L = 0 \quad \Leftrightarrow \quad \begin{cases} \nabla f(\mathbf{w}^*) - \sum_{k=1}^{N} \alpha_k \nabla g_k(\mathbf{w}^*) = 0 \\ g_1(\mathbf{w}^*) = \cdots = g_N(\mathbf{w}^*) = 0 \end{cases}$$

Lagrange Multiplier: Inequality Constraints

- **Consider the following optimization with an inequality constraint:**

$$\text{Minimize } f(x)$$

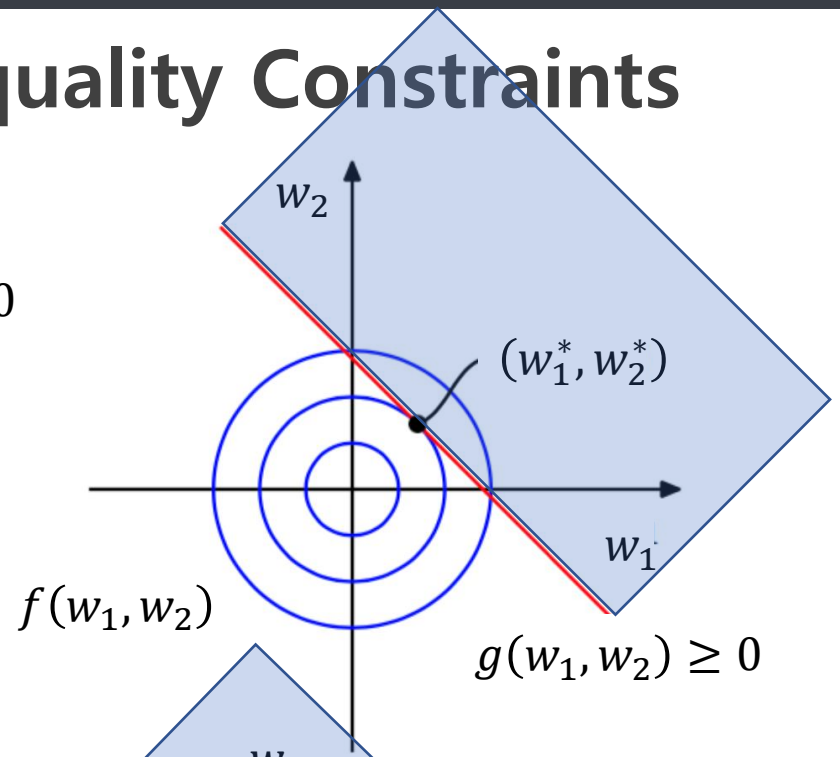
$$\text{Subject to } g(x) \geq 0$$

- **We need to consider two situations:**
  - The constraint is **inactive**: the constraint has no influence
    - In this case, we solve the problem without the constraint
  - The constraint is **active**: we need to consider the constraint
    - Then, the solution is on the tangent points among f and g, so consider the constraint as $g(x) = 0$

Lagrange Multiplier: Inequality Constraints

1. **Active case:**
   1. Optimal point is on g: $g(\mathbf{w}^*) = 0$
   2. Two gradients are parallel:

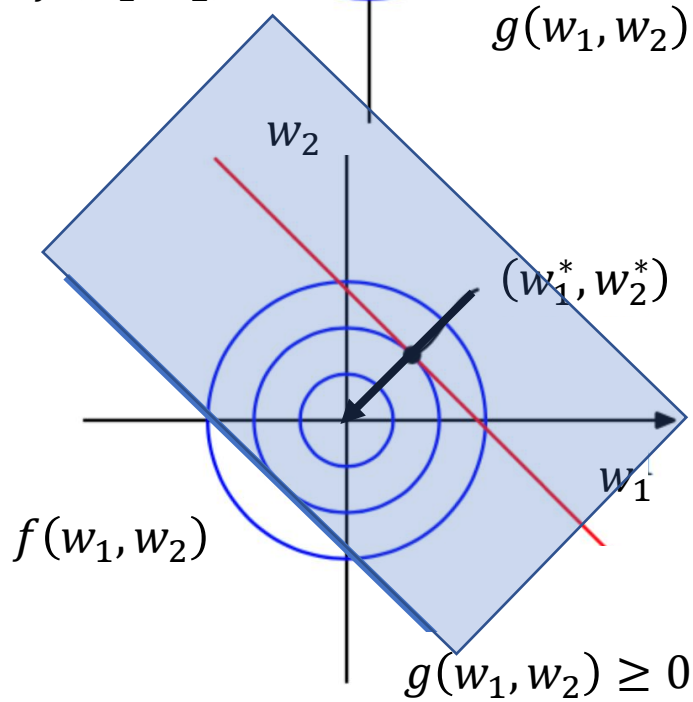
      $\nabla f(\mathbf{w}^*) = \boldsymbol{\alpha} \cdot \nabla g(\mathbf{w}^*)$, $\boldsymbol{\alpha > 0}$



2. **Inactive case:**

   $g(\mathbf{w})$ constraint has no influence

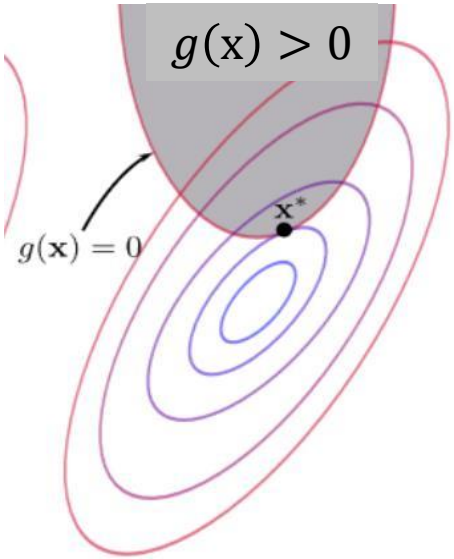
   Optimal point is on $g(\mathbf{w}^*) > 0$



Lagrange Multiplier: Inequality Constraints

1. **Active case:**
   1. Optimal point is on g: $g(\mathbf{w}^*) = 0$
   2. Two gradients are parallel:

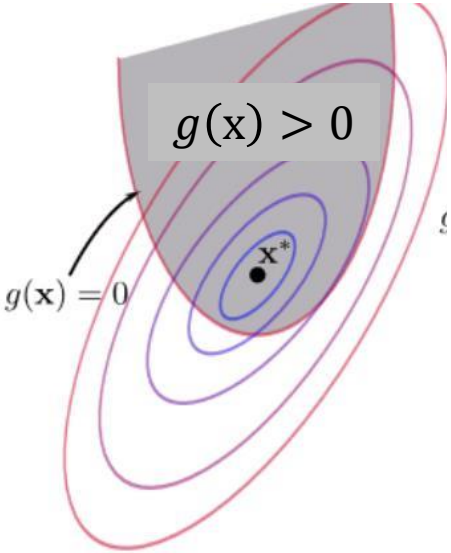
      $\nabla f(\mathbf{w}^*) = \alpha \cdot \nabla g(\mathbf{w}^*)$, $\boldsymbol{\alpha > 0}$



2. **Inactive case:**

   $g(\mathbf{w})$ constraint has no influence

   Optimal point is on $g(\mathbf{w}^*) > 0$



Lagrange Multiplier: Inequality Constraints

- **Lagrange function form:** $L(\mathbf{w}, \alpha) = f(\mathbf{w}) - \alpha g(\mathbf{w})$

  Inactive Constraint : $g(\mathbf{w}^*) > 0,\ \boldsymbol{\alpha = 0},\ \nabla f(\mathbf{w}^*) = 0$

  Active Constraint : $\boldsymbol{g(w^*) = 0},\ \alpha > 0,\ \nabla f(\mathbf{w}^*) = \alpha \nabla g(\mathbf{w}^*)$

- **To consider the above things all together,**
  - Satisfy the **Karush-Kuhn-Tucker (KKT) condition!**

$$\nabla f(\mathbf{w}^*) - \alpha \nabla g(\mathbf{w}^*) = 0, \quad \text{(stationary)}$$
$$\alpha \cdot g(\mathbf{w}^*) = 0, \quad \text{(complementary slackness)}$$
$$\alpha \geq 0, \quad \text{(dual feasibility)}$$
$$g(\mathbf{w}^*) \geq 0 \quad \text{(primal feasibility)}$$

William Karush

Harold W. Kuhn

Albert W. Tucker

[TMI] Harold W. Kuhn and Albert W. Tucker first invented the conditions in 1951. Later William Karush discovered this invention

Lagrange Multiplier: Multiple Inequality Constraints

- **Lagrange function form:**

$$L(\mathbf{w}, \alpha) = f(\mathbf{w}) - \sum_{i=1}^{N} \alpha_i g_i(\mathbf{w})$$

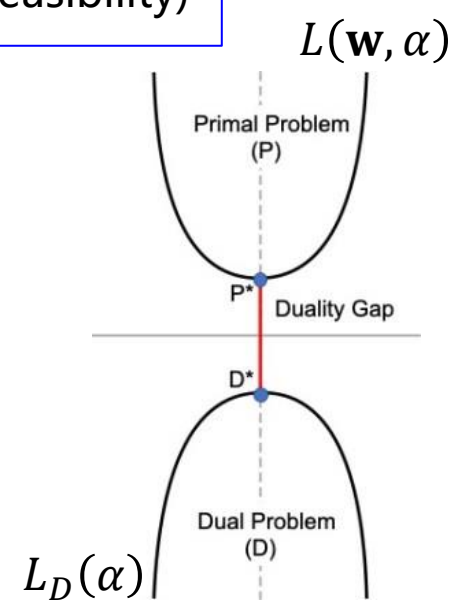
- **Karush-Kuhn-Tucker (KKT) condition**

$\nabla f(\mathbf{w}^*) - \sum_{i=1}^{N} \alpha_i \nabla g_i(\mathbf{w}^*) = 0,$		(stationary)
$\alpha_i g_i(\mathbf{w}^*) = 0,$	$\forall i \in [1, N]$	(complementary slackness)
$\alpha_i \geq 0,$	$\forall i \in [1, N]$	(dual feasibility)
$g_i(\mathbf{w}^*) \geq 0,$	$\forall i \in [1, N]$	(primal feasibility)

- **How to solve?**

1. Solve $\min_{\mathbf{w}} L(\mathbf{w}, \alpha)$, then you will get the **Lagrange Dual function** $L_D(\alpha)$
2. Solve $\max_{\alpha} L_D(\alpha)$ (lagrange dual problem)

why maximize $\boldsymbol{L_D}$? => to reduce the duality gap



Lagrange Multiplier: Final Boss

- **A general constrained minimization problem:**

min $f(x)$

subject to $g_i(x) = c_i$ **for** $i = 1, \ldots, n$ **(Equality constraints)**

$h_j(x) \geq d_j$ **for** $j = 1, \ldots, m$ **(Inequality constraints)**

where $g_i(x) = c_i$ **and** $h_j(x) \geq d_j$

- **Solution: skip…. Not related to SVM**

Application to SVM

- **Defining Lagrange function for SVM**
  - By introducing the Lagrange multiplier a_i

$$\min \quad \frac{1}{2}\|\mathbf{w}\|^2$$

$$s.t. \quad y_i(\mathbf{w}^{\mathrm{T}}\mathbf{x}_i + b) \geq 1, \quad \forall i \in [\mathbf{1}, \boldsymbol{N}]$$

$$\min \quad L_P(\mathbf{w}, b, \alpha_i) = \frac{1}{2}\|\mathbf{w}\|^2 - \sum_{i=1}^{N} \boxed{\alpha_i}\left(y_i\left(\mathbf{w}^{\mathrm{T}}\mathbf{x}_i + b\right) - 1\right)$$

$$\begin{aligned} s.t. \quad & \alpha_i \geq 0, \\ & y_i(\mathbf{w}^{*\mathbf{T}}\mathbf{x}_i + b^*) - 1 \geq 0, \qquad \forall i \in [\mathbf{1}, \boldsymbol{N}] \\ & \alpha_i\left[y_i\left(\mathbf{w}^{*\mathbf{T}}\mathbf{x}_i + b^*\right) - 1\right] = 0, \end{aligned}$$

Thank You